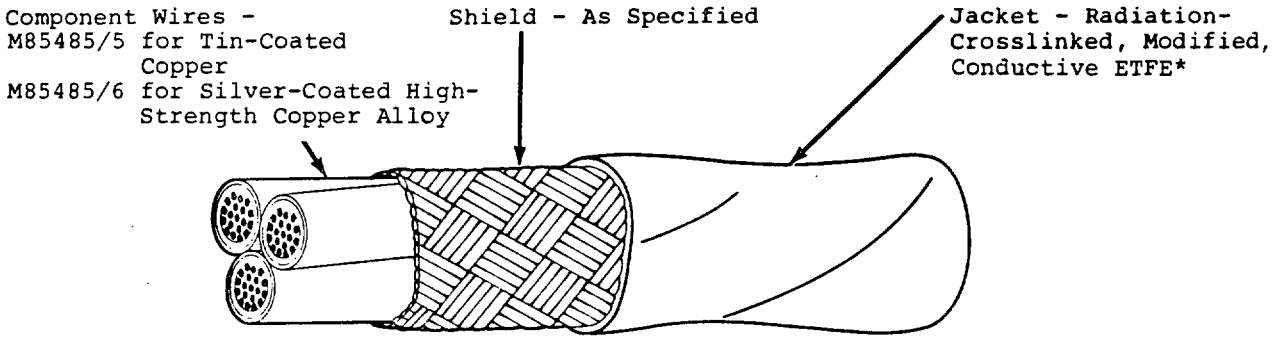
MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, FILTER LINE, SHIELDED, JACKETED, RADIO FREQUENCY ABSORPTIVE, 150°C, 600-VOLT

This specification sheet is approved for use by all departments and agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the issue in effect of Specification MIL-C-85485.



*ETFE - Ethylene-Tetrafluoroethylene Copolymer

FIGURE 1. Typical configuration.

Ratings:

TEMPERATURE RATING: 150°C (302°F) max continuous conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level

Additional requirements:

ATTENUATION (INSERTION LOSS): Consult QPL source for details
BLOCKING: 200° ±3°C (392° ±5°F) for 6 hours
COLOR STRIPING DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
CROSSLINKING PROOF TEST: 7 hours at 300° ±3°C (572° ±5°F)
FLAMMABILITY: 3 seconds (max); 3 inches (max)
IDENTIFICATION DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
IMMERSION: Diameter increase 5% (max); no cracking of jacket
JACKET COLOR: Black
JACKET CONCENTRICITY: 70% (min)
JACKET ELONGATION AND TENSILE STRENGTH: Elongation, 50% (min)
Tensile Strength, 3500 lb f/in^2 (min)
JACKET FLAWS: Not applicable
LIFE CYCLE: 168 hours at 200° ±3°C (392° ±5°F)
LOW TEMPERATURE-COLD BEND: -65° ± 2°C (-85° ±4°F) for 4 hours
SHIELD COVERAGE AND ANGLE: 85% (min) coverage; 18° (min), 40° (max) angle
SHIELD SIZE: 38 AWG round wire strands
SURFACE TRANSFER IMPEDANCE, EFFECTIVE: Consult QPL source for details
THERMAL SHOCK: Oven temperature, 150° ±3°C (302° ±5°F)
Max. change in measurements, 0.125 inch
THERMAL STABILITY: 1250 hours at 180° ±3°C (356° ±5°F)
Voltage Withstand (Dielectric) — No breakdown; Shall meet stop band attenuation requirements for the applicable component wire size and number of components. Consult QPL source for details.
VOLTAGE WITHSTAND (DIELECTRIC); Conductor to shield 1500 volts (rms) 60 Hz
VOLTAGE WITHSTAND (POST-ENVIRONMENTAL): Not applicable

FSC 6145

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TABLE I. Construction details.

Part Number 1/	Conductor Size	Number of Conductors	Jacket Thickness (in.)		Outside Diameter (in.)		Maximum Weight (lbs/1000 ft.)
			(min)	(nom)	(nom.)	(max)	
M85485/8-24*1†	24	1	.0060	.008	.089	.093	9.0
M85485/8-22*1†	22	1	.0060	.008	.095	.100	10.6
M85485/8-20*1†	20	1	.0060	.008	.103	.108	13.1
M85485/8-18*1†	18	1	.0060	.008	.113	.119	16.5
M85485/8-16*1†	16	1	.0060	.008	.119	.125	19.0
M85485/8-14*1†	14	1	.0060	.008	.133	.140	24.9
M85485/8-12*1†	12	1	.0060	.008	.160	.168	36.4
M85485/8-10*1†	10	1	.0060	.008	.184	.193	51.1
M85485/8-24*2†	24	2	.0060	.008	.145	.152	16.1
M85485/8-22*2†	22	2	.0060	.008	.157	.165	19.3
M85485/8-20*2†	20	2	.0060	.008	.173	.182	24.2
M85485/8-18*2†	18	2	.0060	.008	.193	.203	31.0
M85485/8-16*2†	16	2	.0060	.008	.205	.215	35.9
M85485/8-14*2†	14	2	.0070	.009	.235	.247	47.6
M85485/8-24*3†	24	3	.0060	.008	.154	.162	21.2
M85485/8-22*3†	22	3	.0060	.008	.167	.175	25.7
M85485/8-20*3†	20	3	.0060	.008	.184	.193	32.7
M85485/8-18*3†	18	3	.0060	.008	.206	.216	42.5
M85485/8-16*3†	16	3	.0060	.008	.219	.230	49.5
M85485/8-14*3†	14	3	.0070	.009	.251	.264	67.2
M85485/8-24*4†	24	4	.0060	.008	.169	.196	26.4
M85485/8-22*4†	22	4	.0060	.008	.183	.214	32.3
M85485/8-20*4†	20	4	.0060	.008	.202	.237	41.4
M85485/8-18*4†	18	4	.0060	.008	.227	.266	54.2
M85485/8-16*4†	16	4	.0070	.009	.243	.285	64.0
M85485/8-14*4†	14	4	.0070	.009	.277	.326	86.5
M85485/8-24*5†	24	5	.0060	.008	.177	.211	31.4
M85485/8-22*5†	22	5	.0060	.008	.192	.230	38.7
M85485/8-20*5†	20	5	.0060	.008	.213	.255	49.9
M85485/8-18*5†	18	5	.0070	.009	.241	.289	66.3
M85485/8-16*5†	16	5	.0070	.009	.256	.308	77.7
M85485/8-14*5†	14	5	.0070	.009	.292	.352	105.

1/ PART NO.: The asterisks in the part number column shall be replaced by component wire and shield material designators in accordance with MIL-C-85485. The daggers in the part number column shall be replaced by color code designators in accordance with MIL-C-85485.

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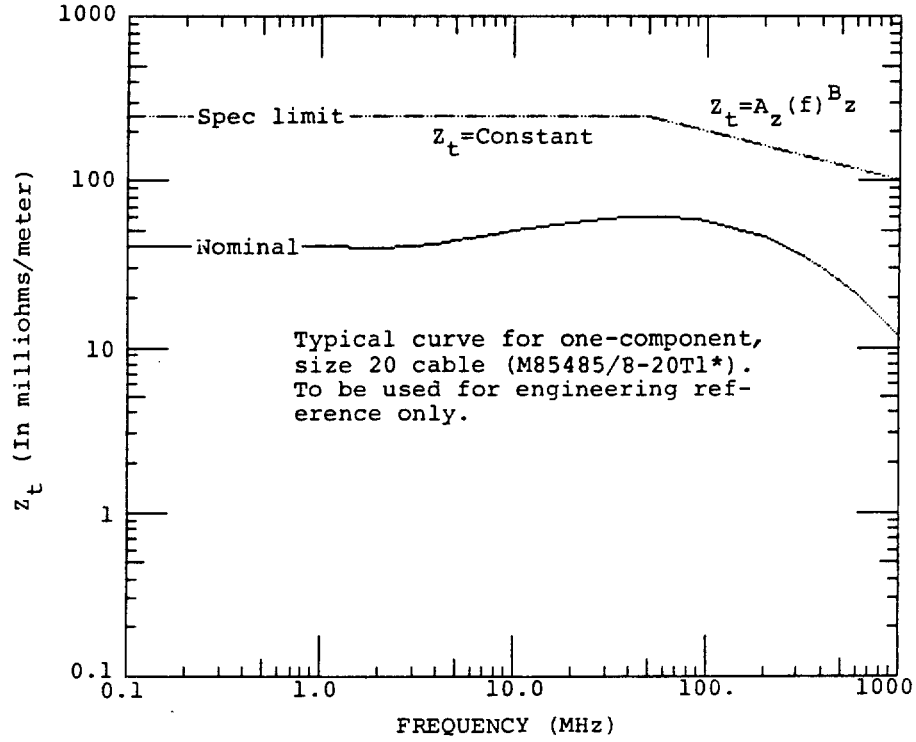


Figure 2. Surface transfer impedance, effective.
ENGINEERING REFERENCE CURVE

Application Note: The component wires of M85485/8 are compatible with all standard connector and contact types, except that the primary jacket diameters of M85485/5-22 component wires exceed the allowable diameter for MIL-C-38999 size 22D contacts. However, these component wires can be terminated to size 22D contacts if the primary jacket and filter layer are stripped in the grommet area, leaving the primary insulation intact.

Qualifying Activity: The activity responsible for this qualified products list is the Naval Air Systems Command, Washington, D.C. 20361.

Custodian:
AF - 85

Preparing activity:
Navy - AS
(Project No. 6145-0840 4)

Review activity:
AF - 19

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